Climbing red crosses: corrupted relay copies at Billaboo and Blackdown Tableland

Investigation 01 · 27 October 2026 · Billaboo AL (535082, rain 5927) and Blackdown Tableland AL (35186, rain 5910), Emerald network · Open: cause narrowed, not confirmed

- **The crosses above the line aren't rain.** Each is a second copy of a real report, damaged as it was relayed. It arrives about a second after a clean copy, with the right count but some upper bits changed.
- **Old ALERT frames have no checksum.** Their few fixed check bits catch damage at the start of a frame but not at the end, where the value's top bits travel. So the damage gets through, and almost always reads high.
- **Blackdown Tableland's odd reset is a real wrap plus the same fault.** Once the counter restarted from 0, its top bit could be forced to 1, which put copies 204.8 mm above the line.
- **Emerald Radar AL is the first place to look:** it's the only repeater both stations go through. Not proven, because FloodNet's receivers don't hear this network.

A clean copy, then a corrupted one a second later

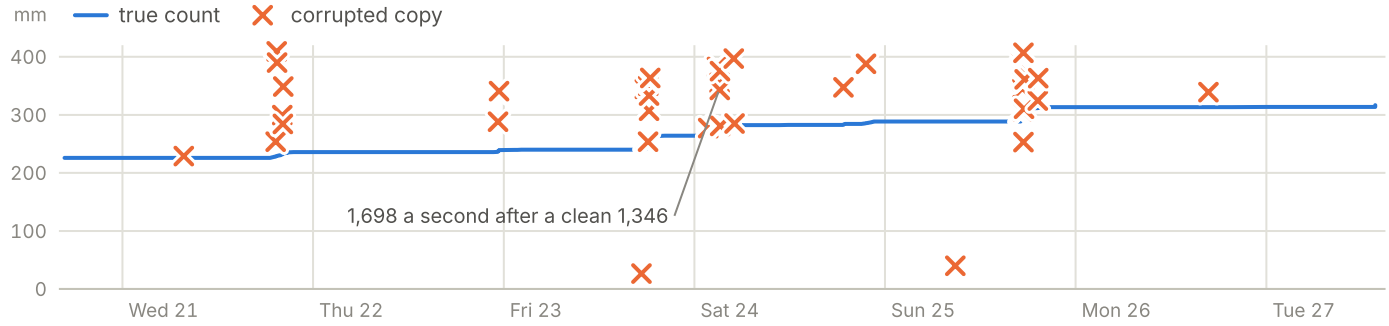


Figure 1. Billaboo AL, 20–27 October. The line is the true count; each cross is one corrupted transmission, usually received as two or more rows. 63 of the 66 sit above the line. Hover a cross for its values.

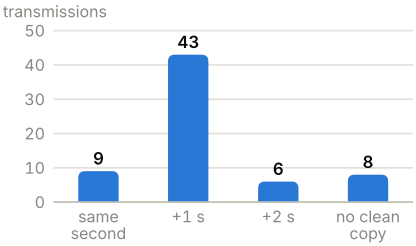


Figure 2. When each corrupted copy arrived, after a clean copy of the same count.

58 of the 66 corrupted transmissions arrived within two seconds of a clean copy of the same count, 43 of them exactly one second after: about one store-and-forward relay hop. Both receive paths in the export got the same wrong value at the same moment, so the damage happened before they split.

Rain multiplies it but doesn't cause it. About the same share of transmissions came back corrupted in heavy rain (14%, 49 of 341), light rain (16%) and the dry three-hourly reports (9%, 5 of 57). The crosses bunch in storms because the gauge transmits on every tip.

Why the damage gets through, and nearly always upward



Figure 3. An ALERT Binary frame is four bytes, each sent lowest bit first between a start bit (0) and a stop bit (1). Bytes 1–2 end in check bits **1 0**; bytes 3–4 end in **1 1**, followed by a stop bit and an idle line that are also 1. Read late, byte 4 shifts every value bit along by one and takes in a 1, and nothing checks it.

Of Billaboo's 66, 32 had one bit forced from 0 to 1 and 10 had two. 14 were the last byte read late, 12 of them by exactly one bit (1,346 became 1,698 three times in twelve minutes on 24 October). Damage early in a frame breaks a **1 0** check, and the frame is dropped; a changed address files the reading under another sensor. So what reaches this sensor was damaged at the end, and it reads high.

Blackdown Tableland: a real wrap, then +1,024 copies

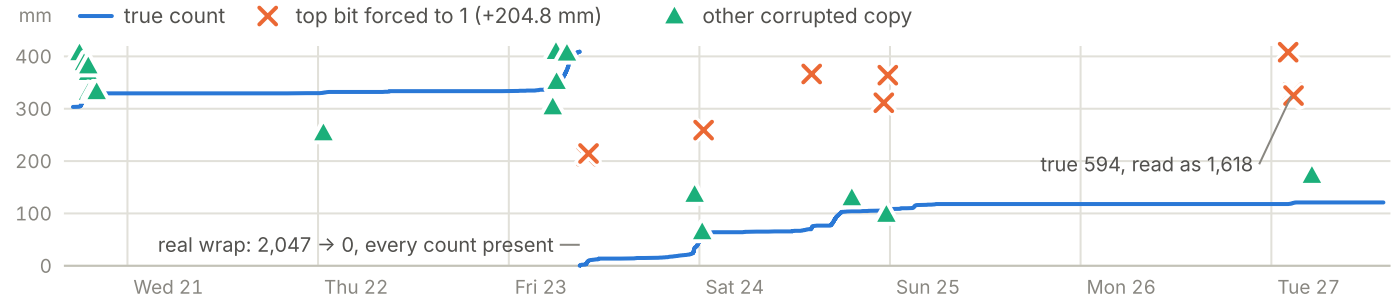


Figure 4. Blackdown Tableland AL, 20–27 October. The wrap on 23 October is real: every count up to 2,047, then 0, 1, 2 …, arrived in order on both paths, graded A.

27 of its 28 corrupted transmissions arrived within two seconds of a clean copy, as at Billaboo. Before the wrap the count was over 1,024, so its top bit was already 1 and couldn't be forced: 0 of 12 corrupted copies changed it. After the wrap the top bit was 0, and 11 of 16 were exactly that bit forced to 1, which put copies like 594 → 1,618 back near the old level.

Across the network: errors bunch at the ends of the frame

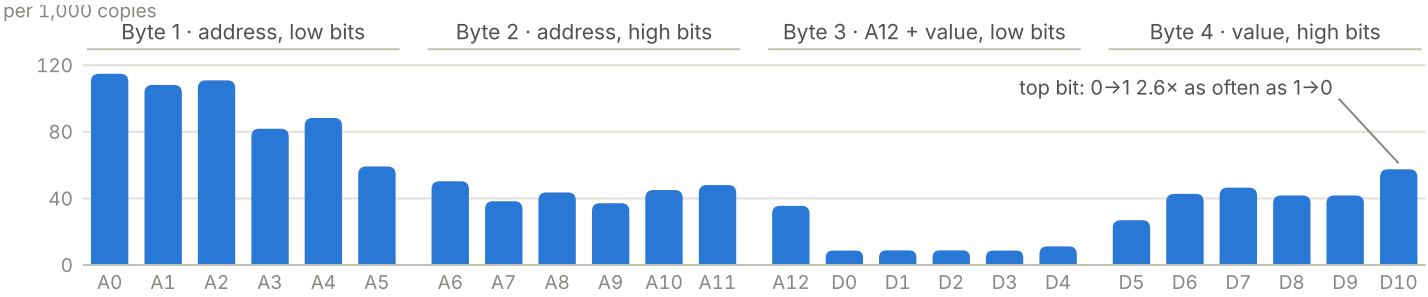


Figure 5. 4,312 corrupted copies heard by Flood-Net's own receivers in south-east Queensland, 7–27 October, from 117 transmitters: how often each bit changed, in the order the bits go on the air. Small changes to low value bits are under-counted, because a reading a few counts out can pass for a real one.

The first and last bytes are hit hardest, which points to timing as a transmission starts and ends rather than to one faulty data line. D5–D9 flip up and down equally, so nothing is stuck high. Only the top bit goes 0→1 2.6× as often as 1→0, as a late read predicts: it hands that bit the check bit after it, which is always 1. Start-of-frame errors fit a frequency synthesiser (PLL) that hasn't settled. That could be a transmitter keying up too fast: those copies came in on carriers 46 Hz from the clean copy's, against 6–14 Hz elsewhere. Or it could be our own decoder's bit-timing loop still locking during a short preamble.

Where it enters

Billaboo is relayed by Mt Mayde (15 km away) and Emerald Radar (94 km); Blackdown Tableland by Roddas Lookout and Emerald Radar, all on 151.5 MHz. At both stations the corrupted copy is one hop behind the clean one, and Emerald Radar is the only repeater on both paths. Both exports also share one receive path's slow, drifting clock, so they reach the Bureau through the same base station.

That narrows it without convicting anyone: Flood-Net's receivers don't hear this network, so carrier fingerprints can't name the sender.

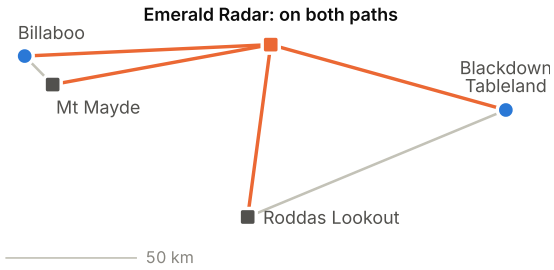


Figure 6. Stations (circles) joined to the repeaters (squares) whose pass ranges include their address.

What we know

Established. These are corrupted relay copies a hop behind a clean one, damaged at the frame's end and almost always high. Blackdown Tableland's wrap is real. Codes DD, PD, ND and AN mark every one of them (3,978 rows checked), so excluding them in **Plots** (the Tyalgum preset) removes the crosses.

Likely. A timing fault at the end of a relayed frame: read a bit late, or the transmitter dropping before its last byte is done.

Open. Which site; and how much start-of-frame error is a transmitter's PLL, how much our decoder.

Next

1. Ask the network maintainer to check Emerald Radar first (receiver, antenna, feeder, key-up and tail timing), then Mt Mayde and Roddas Lookout.
2. Export the same week for Florence Vale AL (relayed by Emerald Radar alone) and Kulumur Range AL (Mt Mayde, then Emerald Radar). Late copies at both point to Emerald Radar; at Kulumur only, to the second hop.
3. Re-decode a batch of our own bursts with a faster bit-timing loop. If first-byte flips drop, part of Figure 5 is ours.

Method: the true count is the longest never-decreasing run of readings in the order received; a copy 16 or more counts (3.2 mm) off it is corrupted; one value repeated within 3 s is one transmission; 8 crosses already flagged by hand (code MM, ARRO's orange triangles) are left out. Data and script: `data/investigations/2026-10-corrupted-relay-copies/`, `tools/investigations/corrupted_relay_copies.py`.